

Experiment No. 23

Write a Prolog Program to suggest Dieting System based on Disease.

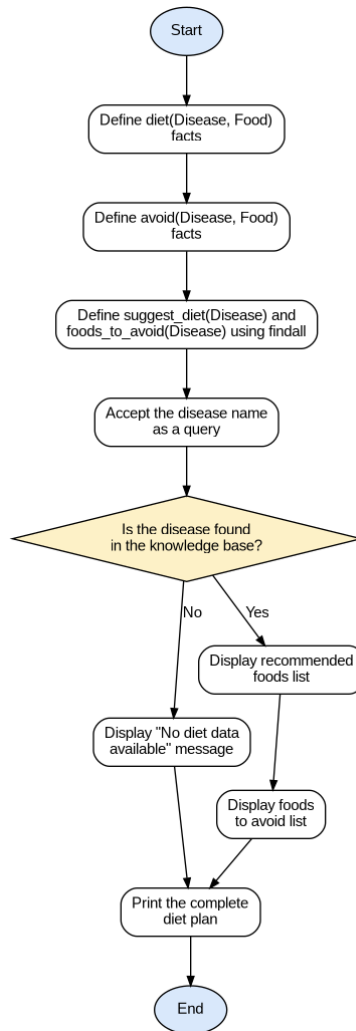
Aim

To write a Prolog program that suggests a suitable diet plan based on a given disease, by defining facts and rules for recommended foods and foods to avoid for each disease.

Algorithm

1. Start.
2. Define facts diet(Disease, FoodItem) that represent food items recommended for each disease.
3. Define facts avoid(Disease, FoodItem) that represent food items to be avoided for each disease.
4. Define the rule suggest_diet(Disease) that uses findall to collect all recommended food items for the given disease.
5. Define the rule foods_to_avoid(Disease) that uses findall to collect all food items to avoid for the given disease.
6. Define the rule diet_plan(Disease) that calls suggest_diet and foods_to_avoid for the given disease.
7. If the disease is found in the knowledge base, display the recommended foods and the foods to avoid.
8. If the disease is not found in the knowledge base, display a message stating that no diet data is available.
9. Accept the disease name as a query to diet_plan.
10. Repeat the query for different diseases.
11. Stop.

Flowchart



Program

```
diet(diabetes, oats).  
diet(diabetes, leafy_vegetables).  
diet(diabetes, whole_grains).  
diet(hypertension, bananas).  
diet(hypertension, low_salt_food).  
diet(hypertension, garlic).  
diet(obesity, salads).  
diet(obesity, fruits).  
diet(obesity, boiled_food).
```

```
avoid(diabetes, sugar).  
avoid(diabetes, white_rice).  
avoid(hypertension, salt).  
avoid(hypertension, pickles).
```

```
avoid(obesity, fried_food).
avoid(obesity, soft_drinks).
```

```
suggest_diet(Disease) :-
    findall(Food, diet(Disease, Food), Foods),
    Foods \= [],
    format("Recommended foods for ~w: ~w~n", [Disease, Foods]).
```

```
foods_to_avoid(Disease) :-
    findall(Food, avoid(Disease, Food), Foods),
    Foods \= [],
    format("Foods to avoid for ~w: ~w~n", [Disease, Foods]).
```

```
diet_plan(Disease) :-
    ( suggest_diet(Disease) -> true
    ; format("No diet data available for ~w~n", [Disease])
    ),
    ( foods_to_avoid(Disease) -> true ; true ).
```

```
main :-
    diet_plan(diabetes),
    diet_plan(hypertension),
    diet_plan(obesity),
    diet_plan(asthma).
```

Result

The program was executed successfully and the output was verified as correct.

?- main.

Recommended foods for diabetes: [oats,leafy_vegetables,whole_grains]

Foods to avoid for diabetes: [sugar,white_rice]

Recommended foods for hypertension: [bananas,low_salt_food,garlic]

Foods to avoid for hypertension: [salt,pickles]

Recommended foods for obesity: [salads,fruits,boiled_food]

Foods to avoid for obesity: [fried_food,soft_drinks]

No diet data available for asthma

true.

Sample Output — Running in VS Code

Screenshot of the program being run in the Visual Studio Code integrated terminal:

dietsystem.pl - AI-Lab - Visual Studio Code

EXPLORER

AI-Lab

sum_n.pl

name_dob.pl

student_teacher.pl

planets.pl

hanoi.pl

birdflight.pl

familytree.pl

dietsystem.pl

dietsystem.pl

1 diet(diabetes, oats).

2 diet(diabetes, leafy_vegetables).

3 diet(diabetes, whole_grains).

4 diet(hypertension, bananas).

5 diet(hypertension, low_salt_food).

6 diet(hypertension, garlic).

7 diet(obesity, salads).

8 diet(obesity, fruits).

9 diet(obesity, boiled_food).

10

11 avoid(diabetes, sugar).

12 avoid(diabetes, white_rice).

13 avoid(hypertension, salt).

14 avoid(hypertension, pickles).

15 avoid(obesity, fried_food).

16 avoid(obesity, soft_drinks).

17

18 suggest_diet(Disease) :-

19 findall(Food, diet(Disease, Food), Foods),

20 Foods \= [],

21 format("Recommended Foods for %s: %s", [Disease, Foods])

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PS C:/Users/Athavan/AI-Lab> swipl dietsystem.pl

Welcome to SWI-Prolog (threaded, 64 bits, version 9.0.4)

?- main.

Recommended foods for diabetes: [oats,leafy_vegetables,whole_grains]

Foods to avoid for diabetes: [sugar,white_rice]

Recommended foods for hypertension: [bananas,low_salt_food,garlic]

Foods to avoid for hypertension: [salt,pickles]

Recommended foods for obesity: [salads,fruits,boiled_food]

Foods to avoid for obesity: [fried_food,soft_drinks]

No diet data available for asthma

true.

?-

Prolog

UTF-8 LF Spaces: 4